

10.19

1.How to build

```
# Compile all src files
Make
```

```
# Compile and run
Make run
```

```
# rm files
Make clean
```

Expected Output
=== Carbon Footprint Report ===

Object	CO2 (kg/year)

Building [name=Office, size=50000 sq ft, energy=850000 kWh/yr]	328,100.00
Building [name=Home, size=2000 sq ft, energy=12000 kWh/yr]	4,632.00
Car [Toyota Camry, 32 miles/yr, 32.0 mpg]	0.00
Car [Ford F-150, 20 miles/yr, 20.0 mpg]	0.00
Bicycle [type=Road, 2000 miles/yr]	68.00
Bicycle [type=Mountain, 1500 miles/yr]	51.00
Bicycle [type=E-Bike, 3000 miles/yr, e-bike]	681.00

TOTAL	333,532.00

2.Function Description

CarbonFootprint.java — (Interface)

```
getCarbonFootprint()
```

Building.java

(All privates, value setted with constructors):

- `name`(String)
- `squareFootage`(double)
- `annualEnergyUsageKwh`(double)

Constructor:

```
Building(String name, double squareFootage, double  
annualEnergyUsageKwh)
```

Getter Method:

- `getName()`
- `getSquareFootage()`
- `getAnnualEnergyUsageKwh()`

`getCarbonFootprint()` Calculations:

```
return annualEnergyUsageKwh * 0.386;
```

Car.java

(All privates, value setted with constructors):

- `make (String)`
- `model (String)`
- `milesPerYear (double)`
- `milesPerGallon (double)`

Constructor:

```
Car(String make, String model, double milesPerYear, double milesPerGallon)
```

Getter Method:

- `getMake()`
- `getModel()`
- `getMilesPerYear()`
- `getMilesPerGallon()`

`getCarbonFootprint()` Calculations:

```
double gallonsPerYear = milesPerYear / milesPerGallon;
```

```
return gallonsPerYear * 8.887;
```

Bicycle.java

(All privates, value setted with constructors):

- `type (String)`
- `milesPerYear (double)`
- `isEbike (boolean)`
- `Kwh (double)`

Constructors:

```
Bicycle(String type, double milesPerYear, boolean isEbike, double kwh)
```

```
Bicycle(String type, double milesPerYear) // Normal Bike
```

Getter Method:

- `getType()`
- `getMilesPerYear()`
- `isEbike()`
- `getKwh()`

`getCarbonFootprint()` Calculations:

```
double footprint = milesPerYear * 0.034;

if (isEbike) {

    footprint += milesPerYear * kwh * 0.386;

}

return footprint;
```